SUNNY RISHY VARDHAN

+1 860 997 9603 | sunnyrishyvardhan@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Software Engineer with an M.S. in Computer Science and hands-on experience in modern C++, Python, Linux, and CMake. Developed concurrent systems, performance benchmarks, and automated tests, with graduate research experience in algorithm design and optimization. Strong foundation in data structures, operating systems, and debugging, with an interest in compiler toolchains and AI software platforms.

Technical Skills

- **Languages:** C++17/20, Python, Java, SQL, TypeScript, Bash
- **C++ and Systems:** STL, Object-Oriented Programming, RAII, Smart Pointers, Multithreading, Atomics, Memory Management
- **Build Tools and Platforms:** CMake, GCC, MSVC, Linux, Windows, Git, Docker
- **Software Engineering:** Data Structures, Algorithms, Debugging, Unit Testing, Performance Profiling, Benchmarking, CI/CD, Code Review
- **AI and Research:** Machine Learning, Model Evaluation, Algorithm Optimization, Experimental Analysis

Experience

Adelphi University Nov 2024 – Present
Graduate Research Assistant in Algorithms and Systems *Garden City, NY*

- Designed a prediction-augmented optimization algorithm for the StrmAds native ad placement problem (WWW 2025, Zhang, Sarpe, Gionis) under Prof. Sixia Chen, extending a 2-approximation greedy approach with a predictions-based framework. Research submitted to ICA3PP 2026.
- Validated the algorithm in Python across 1,500+ synthetic trials and a 1.8M edge real-world YouTube graph, recovering up to 96.7% of a stronger heuristic's performance under high-quality predictions with zero robustness violations against the baseline.
- Profiled inference behavior across the full prediction quality spectrum from ideal to adversarial input, identifying the confidence threshold where the model reverts to worst-case guarantees to guide safe deployment on resource-constrained edge platforms.

express-rate-limit (Open Source Contribution) Jun 2026 – Aug 2026
Merged Contributor | TypeScript, Node.js | PR #661 *Remote*

- Shipped a merged upstream contribution to express-rate-limit (49.8M+ weekly npm downloads) by implementing a new configuration feature across 6 files and landing PR #661 after 3 rounds of design review with 2 core maintainers.
- Diagnosed a suspected production failure by building a live Dockerized Redis reproduction harness, identified the real failure mode differed entirely from the initial hypothesis and documented the root cause before writing any fix.
- Maintained 243 passing tests at 98.6% statement coverage throughout all revisions and kept the CI pipeline green at every iteration.

Projects

High Frequency Trading Simulation Engine *| C++17, CMake, MSVC, GCC, Python | GitHub*

- Achieved approximately 400ns average order matching latency as measured by a dedicated benchmarking harness across multiple load configurations, by implementing a lock free order book where threads submit and cancel orders without a global lock using RAII and smart pointer memory management.
- Validated throughput and latency across five independently testable components as measured by a structured performance profiling report per load configuration, by separating the engine into a matching core, order book, strategy execution layer, backtester and performance monitor.
- Proved end-to-end correctness with zero test failures as measured by a full unit test suite with deterministic synthetic input, by building a Python data generator for reproducible market feeds and maintaining clean CMake Debug and Release build targets throughout.

File Search Engine *| C++17, GCC, Linux | GitHub*

- Built a full-text search engine with zero external dependencies as measured by a single-command build on Linux, by implementing an inverted index that maps each unique term to a sorted posting list using only the C++ standard library, matching the constrained dependency approach required for embedded AI runtime components.
- Delivered relevance-ordered results across multi-term queries as measured by TF-IDF scores computed at query time, by designing a scoring pass that weights per-document term frequency against corpus-wide rarity rather than returning a flat match list.
- Achieved zero test failures across 11 test cases as measured by clean execution covering index construction, single and multi-term queries and ranking accuracy, by keeping all search and scoring logic self-contained so every algorithmic decision is directly traceable.

Adaptive Learning Platform *| Python, FastAPI, Next.js, TypeScript, PostgreSQL, Valkey | Live | GitHub*

- Deployed an AI inference pipeline to production serving real users across multiple active cohorts as measured by live deployment on Vercel and Railway, by building a Python FastAPI backend that adapts content difficulty in real time based on learner performance signals with JWT authenticated role based access control.
- Reduced database load on high-frequency leaderboard reads as measured by improved response latency on ranked score queries, by implementing a Valkey (Redis) cache layer with a cache aside pattern and explicit invalidation on every write.

Education

Adelphi University Aug 2024 – May 2026
Master of Science in Computer Science | GPA: 3.84/4.0 *Garden City, NY*

- Relevant Coursework: Compiler Construction, Advanced Algorithms, Operating Systems, Database Management Systems
- Upsilon Pi Epsilon (UPE), International Honor Society for Computing and Information Disciplines